

RePlastix Innovations: Transforming Plastic Waste into Sustainable Solutions

Salesforce Project Documentation

Student	Information
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Batch	2027 Batch
Project Platform	Salesforce Developer Edition
Project Domain	Plastic Waste Management and Recycled-Product Operations

Project Overview

RePlastix Innovations is a Salesforce-based solution developed to connect plastic collection, recycling-center coordination, recycled-product inventory, customer ordering, and replenishment in one working environment. The implementation uses custom objects, controlled fields, validation rules, role-based security, Scheduled Flow, and Apex to support the complete operational cycle.

Implementation Areas

Area	Covered in this report
Data Management	Objects, tabs, application navigation, fields and relationships
Data Security	Roles, profiles, users and record-level access
Data Configuration	Validation, controlled values and inventory status logic
Automation	Scheduled Flow and Apex inventory processing
Verification	Functional records, notification and successful test coverage

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PHASE 1 — REQUIREMENT ANALYSIS & PLANNING

1.1 Business Objectives

The project begins with a simple operational requirement: information about collected plastic, processing locations, finished products and customer demand should remain connected instead of being maintained as isolated records. Salesforce was selected as the working platform so that the same application can hold the data and execute the rules that act on it.

A second objective is to make inventory activity dependable. The implementation therefore gives the Recycled Product record responsibility for stock information, while Flow and Apex perform the checks and updates that would otherwise require repeated manual work. This creates a traceable path from collection through sales and replenishment.

1.2 Functional Scope

The completed application covers five custom business objects. Together they represent the records needed to follow the movement of material and the resulting products through the organization.

- Plastic Waste — captures collection date, material type, weight, status, location and the associated recycling center.
- Recycling Center — stores facility information such as name, location and processing capacity.
- Recycled Product — keeps product identity, available stock, threshold, price and the calculated stock condition.
- Order — records the customer demand, selected recycled product, quantity and delivery information.
- Restock Request — represents a shortage and carries the requested quantity and current replenishment status.
- Automation — performs scheduled low-stock checking, order-driven inventory processing, shortage creation, approved-restock processing and notification.

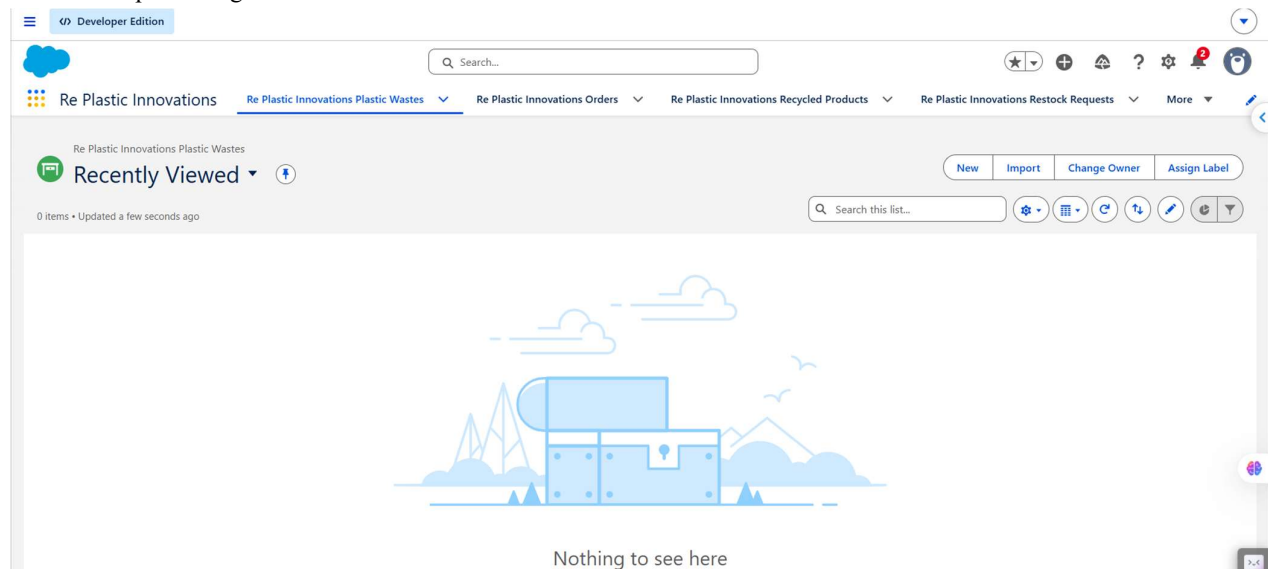


Figure 1: Re Plastic Innovations Lightning application and project navigation

1.3 Stakeholder Mapping

The solution separates responsibilities according to the way the organization would operate. Management requires wider visibility, while collection, sales and warehouse users work mainly with the records belonging to their activities. Salesforce roles provide the organizational hierarchy and profiles supply the baseline permissions for these users.

Stakeholder / Role	Primary Responsibility	Relevant Access
CEO	Overall oversight	Higher-level record visibility
Recycling Manager	Plastic-waste operations	Plastic Waste create/read and shared waste records
Sales Representative	Customer orders and products	Order and Account create/read; product read
Warehouse Supervisor	Inventory and restocking	Custom-object create/read/edit; restock visibility



SETUP

Roles

Creating the Role Hierarchy

You can build on the existing role hierarchy shown on this page. To insert a new role, click **Add Role**.

Your Organization's Role Hierarchy

[Collapse All](#) [Expand All](#)

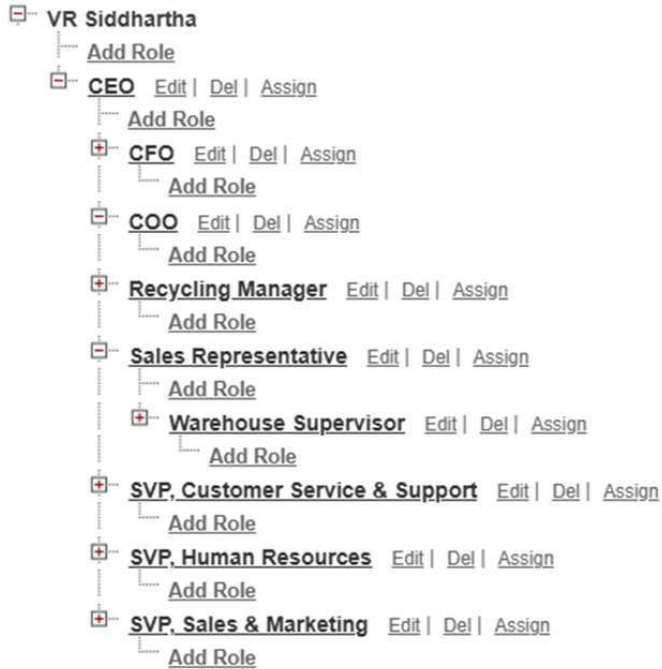


Figure 2: Configured Salesforce role hierarchy

1.4 Execution Roadmap

Implementation was carried out from the data layer upward. The objects and fields were established first, followed by the application interface, business rules, automation, security and testing. This sequence makes each stage depend on components that have already been defined.

- Identify the business records and define the five custom objects.
- Add fields, relationships, picklist values and navigation tabs.
- Assemble the Re Plastic Innovations Lightning application.
- Add the formula and validation rules that protect data quality.
- Configure the scheduled Flow for low-stock monitoring.
- Develop the Apex inventory processing and event triggers.
- Apply profiles, roles, organization-wide defaults and sharing rules.
- Create representative records and execute the planned test scenarios.

PHASE 2 — SALESFORCE DEVELOPMENT – BACKEND & CONFIGURATIONS

2.1 Data Architecture

The backend model divides the business into records that have different meanings and lifecycles. Plastic Waste describes incoming material, Recycling Center describes the facility, Recycled Product represents sellable inventory, Order records demand, and Restock Request records replenishment. Lookup relationships connect these records without duplicating their information.

Object	Important Fields	Relationship
Plastic Waste	Weight, Type, Collection Date, Status, Location	Lookup to Recycling Center
Recycling Center	Location, Capacity	Referenced by Plastic Waste
Recycled Product	Stock Level, Threshold, Price	Referenced by Order and Restock Request
Order	Customer, Product, Quantity, Delivery Date	Lookup to Account and Recycled Product
Restock Request	Product, Requested Quantity, Status	Lookup to Recycled Product

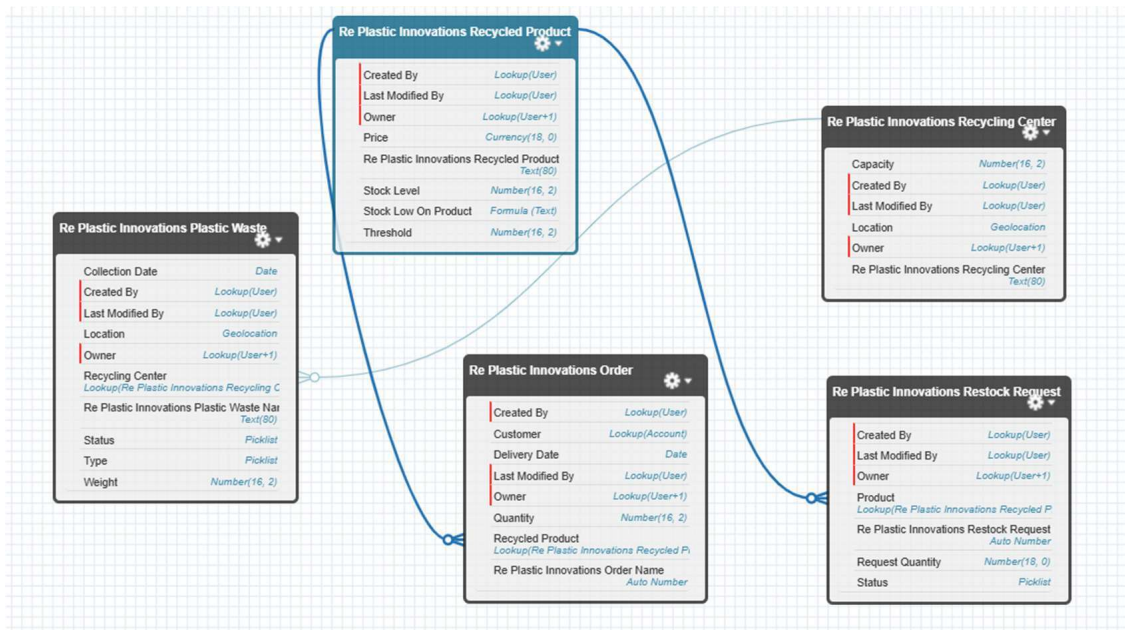


Figure 3: RePlastix Innovations object relationship structure

2.2 Automation Logic

Automation is divided according to when the action must occur. The scheduled Flow handles a recurring inventory review, whereas Apex responds directly to record transactions. This separation keeps the time-based check independent from the order and restock events that modify stock.

Automation	When it runs	Main action
Scheduled Flow — Stock Level Is Low	Daily at 6:00 AM	Compares Stock Level with Threshold and creates a high-priority Task when stock is low
UpdateStockAfterOrder	After Order insert	Uses InventoryManager to reduce stock or create a shortage Restock Request
UpdateStockAfterRestockApproval	After Restock Request update	Recognizes Pending → Approved, increases stock and calls the email helper

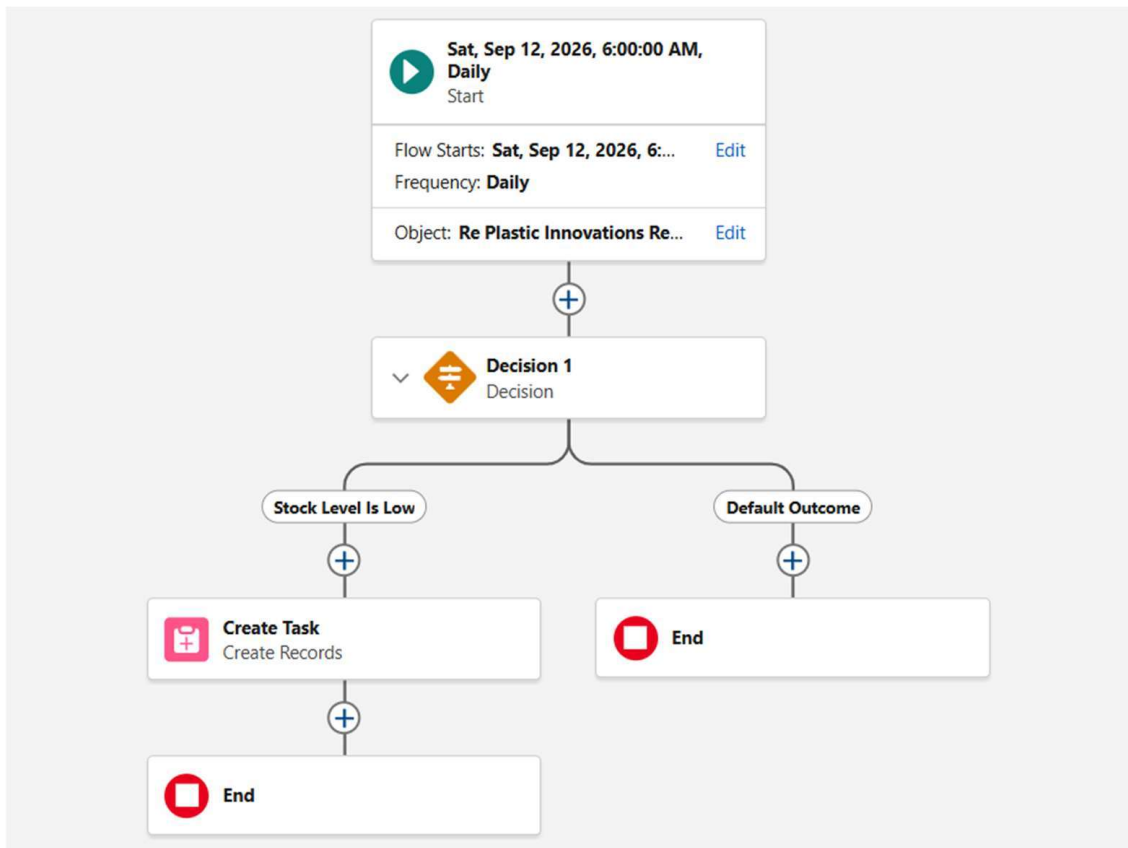


Figure 4: Scheduled low-stock Flow configuration

Re Plastic Innovations Restock Request
Req-0001

Related Details

Re Plastic Innovations Restock Request Name	Req-0001	Owner	Zaheeruddin Shaik
Product	Recycled Plastic Test Product		
Request Quantity	100		
Status	Pending		
Created By	Zaheeruddin Shaik, 9/16/2026, 6:18 PM	Last Modified By	Zaheeruddin Shaik, 9/16/2026, 6:18 PM

Figure 5: Restock transaction evidence used with the inventory workflow

2.3 Business Rules

The project keeps straightforward restrictions in Salesforce configuration and uses Apex where a transaction must affect more than one related record. The formula communicates inventory status, validation rules reject unsuitable inputs, Flow creates the scheduled follow-up, and Apex performs the stock transactions.

- Stock status shows 'Low Stock - Restock Needed' below the configured threshold and 'Sufficient Stock' otherwise.
- Order Quantity is required to be greater than zero.
- Plastic Waste Collection Date is restricted from accepting a future date.
- An order with enough stock subtracts its quantity from the related Recycled Product.
- An order that exceeds available stock creates a Pending Restock Request for the shortage.
- When a Pending request becomes Approved, the requested quantity is added to product stock and the notification helper is called.



Re Plastic Innovations

Re Plastic Innovations Plastic Wastes

Re Plastic Innovations Orders

Re Plastic Innovations Recycled Products

Re Plastic In

Re Plastic Innovations Recycled Product

 Recycled Plastic Test Product

Related

Details

Re Plastic Innovations Recycled Product Name

Recycled Plastic Test Product

Owner

[Zaheeruddin Shaik](#)

Stock Level

150.00

Threshold

100.00

Price

\$1,000

Stock Low On Product

Sufficient Stock

Created By

[Zaheeruddin Shaik](#), 9/16/2026, 6:15 PM

Last Modified By

[Zaheeruddin Shaik](#), 9/16/2026, 6:20 PM

Figure 6: Recycled Product record showing inventory fields and stock-status information

PHASE 3 — UI/UX DEVELOPMENT & CUSTOMIZATION

3.1 Interface Design

The Re Plastic Innovations Lightning application acts as the user-facing layer of the project. The important records are exposed through application navigation so that a user can work with the solution without entering Setup for normal operational tasks. The interface therefore presents the business process through records and tabs rather than configuration menus.

The selected navigation areas correspond to the five custom objects. This gives the application a consistent working space for collection, facility information, inventory, ordering and replenishment.

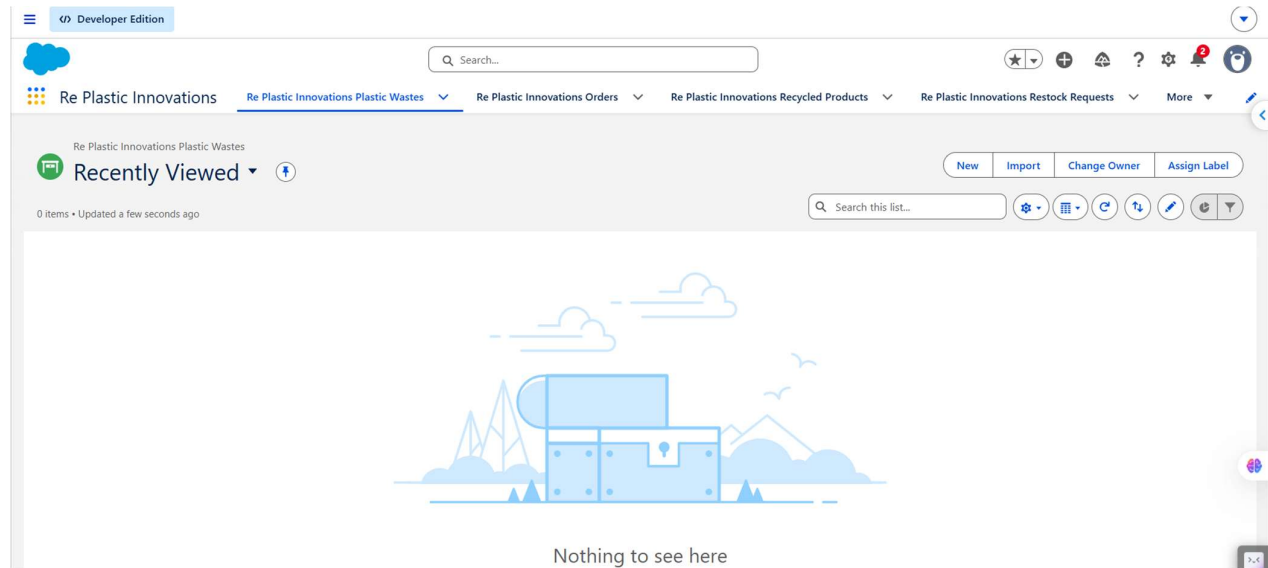


Figure 7: Re Plastic Innovations Lightning application interface

3.2 Navigation Flow

Navigation follows the movement of information through the business process. A waste record is associated with a recycling center; finished products hold the inventory; an order consumes that inventory; and a restock request provides the path back to the product when additional stock is needed.

Starting Point	Next Record / Action	Purpose
Plastic Waste	Recycling Center	Connect collected material with its processing facility
Recycled Product	Order	Use available inventory to satisfy customer demand
Order	Restock Request	Record the shortage when stock is not sufficient
Restock Request	Recycled Product	Reflect approved replenishment in inventory

3.3 Usability

Usability is supported by predictable labels, controlled picklist choices, meaningful record fields and calculated status information. A user can identify the condition of a product from the stock-status field without manually comparing the stock and threshold values.

Validation rules also provide a direct safeguard at data entry. Together with the automated inventory actions, these features reduce the amount of repetitive checking required from operational users and make the system easier to demonstrate and maintain.

PHASE 4 — DATA MIGRATION, TESTING & SECURITY

4.1 Data Handling

Data handling is centered on structured fields rather than free-form descriptions. Dates identify collection and delivery events, numbers represent weights, quantities and stock levels, currency stores product price, picklists control categories and states, and lookup fields connect records that belong to the same business transaction.

The use of consistent field types also gives later automation reliable inputs. For example, the stock formula can compare numeric values, while Apex can retrieve the related product from an Order or Restock Request and update the correct inventory record.

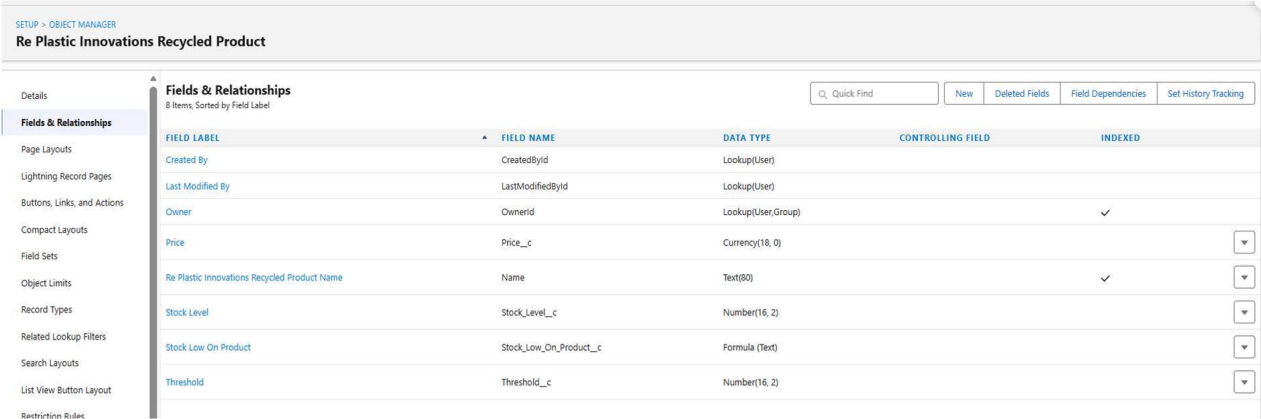


Figure 9: Recycled Product field configuration

4.2 Access Control

Security was implemented as multiple layers rather than one permission setting. Profiles establish what an authenticated user can do with the objects, roles establish organizational position, organization-wide defaults define the baseline record visibility, and sharing rules extend access where collaboration is required.

Control	Purpose	Implemented Example
Profiles	Object and functional permissions	Project profiles provide the required Create, Read, Edit and Delete access
Roles	Hierarchy and sharing context	CEO, Recycling Manager, Sales Representative and Warehouse Supervisor
OWD	Default record visibility	Relevant custom objects configured with private visibility
Sharing Rules	Additional record access	Selected access paths between management and operational roles

4.3 QA Testing

Testing concentrates on the inventory events that determine whether the application behaves correctly in normal and shortage conditions. The test data includes a Recycled Product with an available stock value and a lower threshold, after which the order and replenishment scenarios are exercised.

Test Scenario	Expected Result	Observed Result
Order quantity within available stock	Stock decreases by ordered quantity	Passed
Order quantity exceeds available stock	Pending Restock Request records the shortage	Passed
Restock Request changes to Approved	Product stock increases by requested quantity	Passed

Status	Test Run	Enqueued Time	Duration	Failures	Total
✓	TestRun @ 7:22:57 am			0	4
✓	TestRun @ 7:23:18 am			0	4
✓	TestRun @ 7:27:36 am			0	5

Overall Code Coverage		
Class	Percent	Lines
Overall	100%	
EmailNotificationHelper	100%	16/16
InventoryManager	100%	47/47
UpdateStockAfterOrder	100%	1/1
UpdateStockAfterRestockApproval	100%	11/11

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UpdateStockAfterRestockApproval	100%	11/11

Figure 11: Apex test execution and code-coverage result

4.4 Data Integrity

Data integrity is maintained by combining controlled values, validation rules, formula logic and programmatic processing. The collection-date rule prevents a future operational date, the order rule prevents a zero or negative quantity, and the stock formula keeps the displayed inventory condition synchronized with its source values.

The inventory triggers also use the business event that caused the change. In particular, the restock logic processes the transition into Approved so that a later edit to an already approved request does not represent another approval event.

SETUP > OBJECT MANAGER

Re Plastic Innovations Plastic Waste

Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets

Re Plastic Innovations Plastic Waste Validation Rule
[Back to Re Plastic Innovations Plastic Waste](#)

Validation Rule Detail Edit Clone

Rule Name	Future_Date_Collection	Active	✓
Error Condition Formula	Collection_Date__c > TODAY()		
Error Message	Collection Date cannot be in the future.	Error Location	Collection Date
Description			
Created By	Zaheeruddin Shaik, 9/12/2026, 9:22 AM	Modified By	Zaheeruddin Shaik, 9/12/2026, 9:22 AM

Edit Clone

Figure 12: Collection-date validation configuration

SETUP > OBJECT MANAGER

Re Plastic Innovations Order

Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types

Re Plastic Innovations Order Validation Rule
[Back to Re Plastic Innovations Order](#)

Validation Rule Detail Edit Clone

Rule Name	Check_Quantity_Not_Zero	Active	✓
Error Condition Formula	Quantity__c <= 0		
Error Message	Quantity must be greater than zero.	Error Location	Quantity
Description			
Created By	Zaheeruddin Shaik, 9/12/2026, 9:21 AM	Modified By	Zaheeruddin Shaik, 9/12/2026, 9:21 AM

Edit Clone

Figure 13: Order quantity validation configuration

PHASE 5 — DEPLOYMENT, DOCUMENTATION & FINAL PRESENTATION

5.1 Troubleshooting

The final implementation was checked across the main dependency points: object and field references, lookup connections, Flow configuration, trigger execution, stock calculations, notification behavior and test coverage. Each component was reviewed in the context of the business event it supports so that the final workflow remains consistent from order creation through replenishment.

The Apex components are separated by responsibility. InventoryManager contains the reusable stock-processing logic, while the order and restock triggers invoke that logic at their respective events. EmailNotificationHelper handles the notification step after an approved replenishment.

5.2 Adherence to Timelines

The work was organized in stages so that later tasks could use stable outputs from earlier tasks. Data modeling came first, followed by the application layer, business rules, automation, security and verification. This approach also made it easier to demonstrate each feature as a completed part of the overall solution.

5.3 Innovation

The implementation combines Salesforce configuration with custom Apex to represent an end-to-end recycled-product inventory process. The distinctive part of the workflow is the connection between demand and replenishment: an order can either consume available stock or create a shortage record, and an approved restock can then update the same product inventory and trigger a notification.

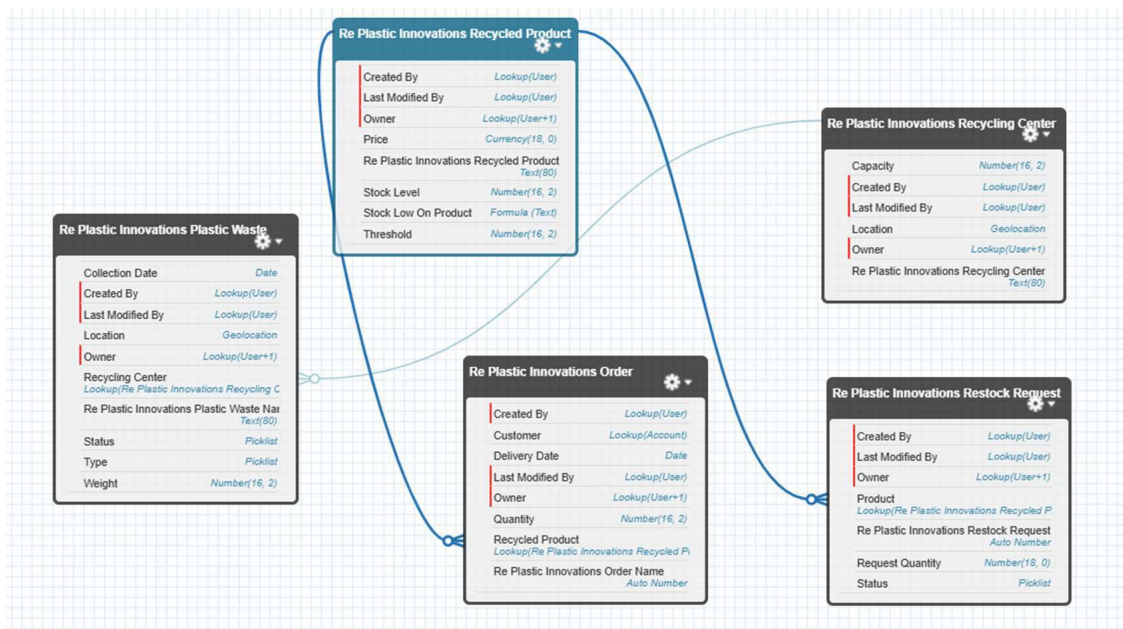


Figure 14: End-to-end data relationships supporting the RePlastix Innovations process

5.4 Document Functional Overview

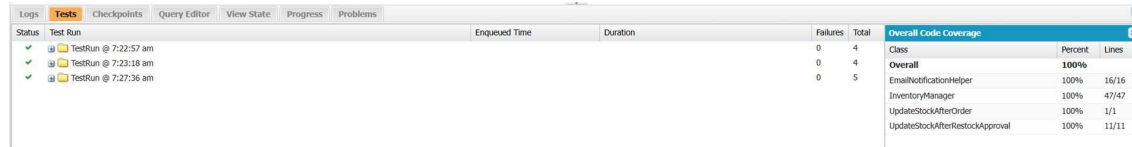
From a user perspective, the application supports collection records, recycling-center information, finished-product inventory, customer orders and replenishment requests. These areas are accessible from the Lightning application and are connected through lookup relationships.

The main operational sequence is straightforward: a product is available for sale, an order is entered, the system checks inventory, and the result determines whether stock is reduced or a Pending Restock Request is created. After the request is approved, inventory is increased and a notification is generated.

5.5 Document Technical Blueprint

The technical design has four cooperating layers. The data layer contains five custom objects and their relationships. The interface layer contains the Lightning application and custom tabs. The rule layer contains the formula and validation rules. The automation layer contains Scheduled Flow and Apex classes/triggers for the transaction-driven inventory process.

Layer	Implemented Components
Data	Plastic Waste, Recycling Center, Recycled Product, Order, Restock Request and Account lookup
UI	Re Plastic Innovations Lightning App and custom tabs
Declarative Logic	Stock status formula, two validation rules and scheduled low-stock Flow
Programmatic Logic	InventoryManager, UpdateStockAfterOrder, UpdateStockAfterRestockApproval and EmailNotificationHelper
Security	Profiles, roles, private OWD and sharing rules
Testing	InventoryManagerTest covering the core inventory scenarios



Status	Test Run	Enqueued Time	Duration	Failures	Total
✓	TestRun @ 7:22:57 am			0	4
✓	TestRun @ 7:23:18 am			0	4
✓	TestRun @ 7:27:36 am			0	5

Overall Code Coverage		
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Overall	100%	
EmailNotificationHelper	100%	16/16
InventoryManager	100%	47/47
UpdateStockAfterOrder	100%	1/1
UpdateStockAfterRestockApproval	100%	11/11

Figure 15: Salesforce development configuration supporting the technical implementation

5.6 Document Setup Manual

A reproducible setup can be followed in the same dependency order used during implementation:

- Create Plastic Waste, Recycling Center, Recycled Product, Order and Restock Request.
- Add the required fields, picklist values and lookup relationships.
- Create the corresponding custom tabs and configure the Re Plastic Innovations Lightning application.
- Add the stock-status formula and the collection-date and order-quantity validation rules.
- Build and activate the scheduled low-stock Flow.
- Create InventoryManager, UpdateStockAfterOrder, UpdateStockAfterRestockApproval and EmailNotificationHelper.
- Configure profiles, roles, organization-wide defaults and sharing rules.
- Create representative records and execute the Apex tests to verify the completed workflow.

5.7 Project Demo Video Planning

The demonstration should concentrate on the working business process and use configuration screens only where they help explain a feature. A short sequence is enough to show the application, inventory, order processing, replenishment and verification without turning the video into a step-by-step Setup tutorial.

- Introduce the student, project title and problem being addressed.
- Open the Re Plastic Innovations application and show the navigation.
- Open a Recycled Product record and point out Stock Level, Threshold and stock status.
- Create or show an order that can be satisfied from available stock and demonstrate the inventory change.
- Show the shortage path and the resulting Pending Restock Request.
- Change the request to Approved and show the updated product stock and notification.
- Briefly show the Scheduled Flow and Apex test result as technical evidence.
- Close with the end-to-end outcome of the system.

5.8 Visual Demonstration

The visual evidence should be selected so that every important claim in the report has a corresponding Salesforce screen. The application screenshot establishes the user workspace, the field and relationship screens establish the data model, the role screen establishes the security hierarchy, the Flow screen proves scheduled automation, and the test screen demonstrates verification.

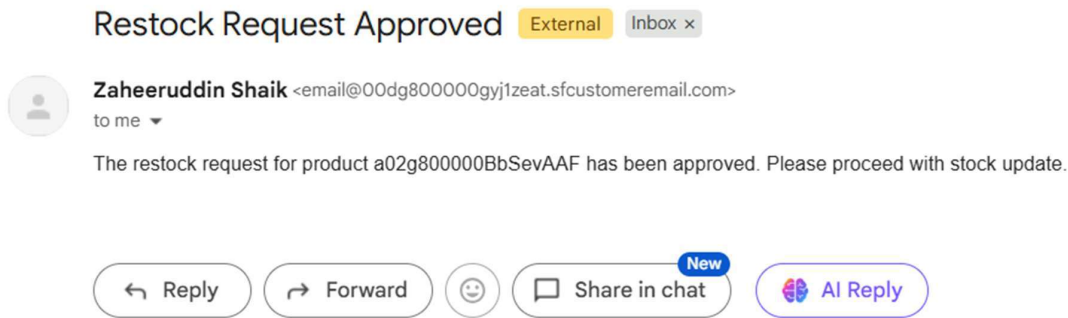


Figure 16: Restock approval notification evidence

5.9 Explanation Clarity

The project can be explained in a consistent order: first describe the operational problem, then introduce the Salesforce solution, followed by the objects and relationships, rules, automation and security. The live order-to-restock scenario should then connect these individual components into one understandable process.

When describing the replenishment feature, the implementation should be presented as a status-driven workflow. A shortage produces a Pending Restock Request, the request is changed to Approved, and the after-update trigger processes that transition. This accurately reflects the configuration demonstrated by the project.

5.10 Realism & Quality

The project models a realistic business situation in which the availability of recycled products is affected by customer demand. It includes the supporting records needed by collection, operations, sales and warehouse activities, while automation handles the repeated inventory decisions that would otherwise require manual follow-up.

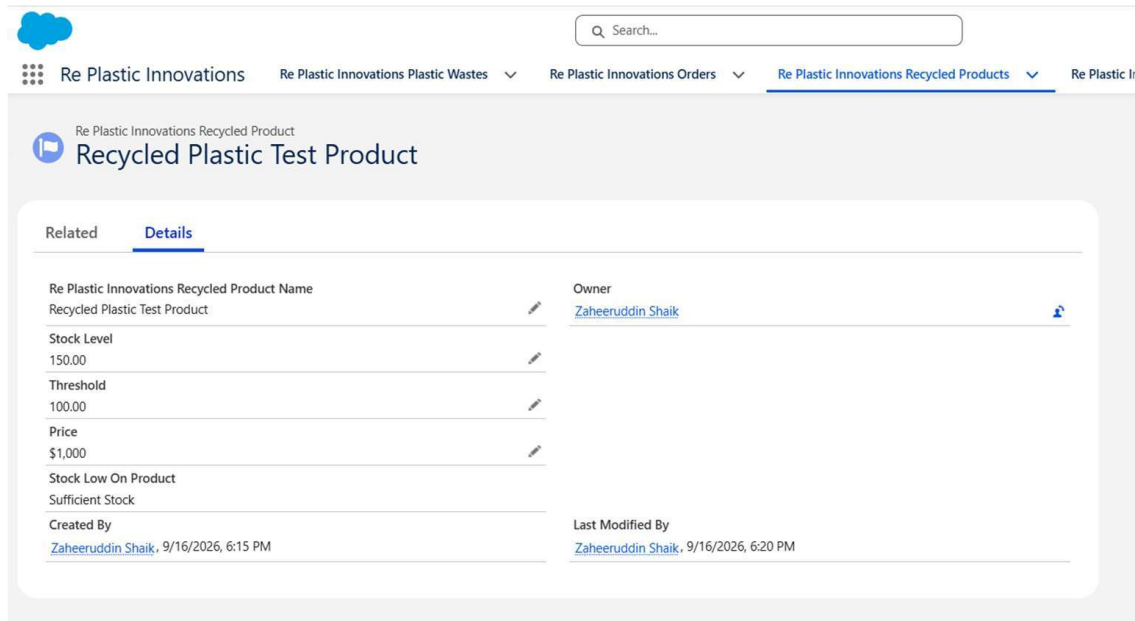


Figure 17: Recycled Product record representing the operational inventory view

5.11 Scalability & Future Plan

The current design provides a base for additional Salesforce features. Future work could introduce management dashboards, richer reports, more detailed recycling stages, expanded notification options, supplier or vendor records, and additional inventory analytics. Because the present solution keeps business entities and automation responsibilities separated, these extensions can be added without changing the purpose of the existing core records.

END OF PROJECT DOCUMENTATION

RePlastix Innovations: Transforming Plastic Waste into Sustainable Solutions